

Assignment 8

Use sub-query to solve following problems.

Note: To solve below queries, use “sales” database

1.

Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
kd1_99523_sandhya1> select * from orders where cnum=(select cnum from customers where cname='Cisneros');
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3001 | 18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

7.

Extract all the orders of Motika.

```
kd1_99523_sandhya1> select * from orders where snum=(select snum from salespeople where sname='motika');
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

8.

Find all the orders of salespeople servicing customers in London.

```
kd1_99523_sandhya1> select * from orders where snum IN (select snum from customers where city='london');
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

6.

Write a query that selects all orders for amounts greater than any for the customers in London.

```
kd1_99523_sandhya1> select * from orders where amt > ANY(select amt from orders where cnum IN (select cnum from customers where city='London'));
```

onum	amt	odate	cnum	snum
3002	1900.10	1990-10-03	2007	1004
3005	5160.45	1990-10-03	2003	1002
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3008	4723.00	1990-10-04	2006	1001
3011	9891.88	1990-10-04	2006	1001

6 rows in set (0.00 sec)

4.

Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of serres salesperson).

```
kd1_99523_sandhya1> select * from customers where rating >= ANY (select rating from customers where snum in (select snum from salespeople where sname='Serres'));
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

4 rows in set (0.00 sec)

11.

Select customers who have a greater rating than any other customer in Rome.

```
kd1_99523_sandhya1> select * from customers where rating > ANY (select rating from customers where city='Rome');
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

4 rows in set (0.00 sec)

12.

Select all orders that had amounts that were greater than at least one of the orders from '1990-10-04'.

```
kd1_99523_sandhya1> select * from orders where amt > ANY(select amt from orders where odate='1990-10-04');
```

onum	amt	odate	cnum	snum
3003	767.19	1990-10-03	2001	1001
3002	1900.10	1990-10-03	2007	1004
3005	5160.45	1990-10-03	2003	1002
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3008	4723.00	1990-10-04	2006	1001
3010	309.95	1990-10-04	2004	1002
3011	9891.88	1990-10-04	2006	1001

8 rows in set (0.00 sec)

13.

Find all orders with amounts smaller than any amount for a customer in San Jose.

```
kd1_99523_sandhya1> select * from orders o where o.amt < ANY(select amt from orders where cnum IN (select cnum from customers where city='San Jose'));
```

onum	amt	odate	cnum	snum
3001	18.69	1990-10-03	2008	1007
3003	767.19	1990-10-03	2001	1001
3002	1900.10	1990-10-03	2007	1004
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3007	75.75	1990-10-04	2004	1002
3008	4723.00	1990-10-04	2006	1001
3010	309.95	1990-10-04	2004	1002

8 rows in set (0.00 sec)

2.

Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

```
kd1_99523_sandhya1> select o.amt, c.cname, c.rating from orders o JOIN customers c ON o.cnum=c.cnum where amt > (select avg(o.amt) from orders o);
```

amt	cname	rating
5160.45	Liu	200
9891.88	Clemens	100
4723.00	Clemens	100

3 rows in set (0.00 sec)

14.

Select those customers whose rating are higher than every customer in London.

```
kd1_99523_sandhya1> select * from customers where rating >ALL(select rating
from customers where city='london');
+-----+-----+-----+-----+
| cnum | cname   | city   | rating | snum |
+-----+-----+-----+-----+
| 2002 | Giovanni | Rome   | 200    | 1003 |
| 2003 | Liu      | San Jose | 200    | 1002 |
| 2004 | Grass    | Berlin | 300    | 1002 |
| 2008 | Cisneros | San Jose | 300    | 1007 |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

5.

Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

```
kd1_99523_sandhya1> select * from salespeople where city!=ALL(select city f
om customers);
+-----+-----+-----+-----+
| snum | sname   | city   | comm |
+-----+-----+-----+-----+
| 1007 | Rifkin  | Barcelona | 0.15 |
| 1003 | Axelrod | New York | 0.10 |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

9.

Find names and numbers of all salesperson who have more than one customer.

```
kd1_99523_sandhya1> select sname,snum from salespeople where snum IN (selec
t snum from customers group by snum having count(*) >1);
+-----+-----+
| sname | snum |
+-----+-----+
| Peel  | 1001 |
| Serres | 1002 |
+-----+-----+
2 rows in set (0.00 sec)
```

10.

Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
kd1_99523_sandhya1> select snum, sname , city, (select count(*) from custome
rs c where s.snum=c.snum) as c_count from salespeople s where (select count(
*) from customers c where s.snum=c.snum)>1;
+-----+-----+-----+-----+
| snum | sname   | city   | c_count |
+-----+-----+-----+-----+
| 1001 | Peel    | London | 2        |
| 1002 | Serres  | San Jose | 2        |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

3.

Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.

```
kd1_99523_sandhya1> SELECT snum, SUM(amt)
-> FROM orders
-> GROUP BY snum
-> HAVING SUM(amt) > (SELECT MAX(amt) FROM orders);
```

snum	SUM(amt)
1001	15382.07

1 row in set (0.00 sec)